

Continual Learning for Non-Stationary Smart Grid Load Forecasting: An Elastic Weight Consolidation with Experience Replay Framework

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Abstract

Short-term load forecasting (STLF) models deployed in smart grid control centers must operate continuously under non-stationary conditions driven by seasonal demand cycles, evolving consumption patterns, and increasing renewable energy penetration. Conventional practice of periodic model retraining introduces operational downtime and computational overhead, while naive sequential fine-tuning of neural network models leads to catastrophic forgetting of previously learned patterns. This paper proposes EWC-IXER, a continual learning framework that combines Elastic Weight Consolidation (EWC) with importance-weighted experience replay (IXER) to enable stable sequential learning across distribution-shifted tasks without revisiting raw historical data. The framework additionally integrates conformal prediction-based metacognitive monitoring for adaptive regularization and Bayesian change point detection for automatic task boundary identification. We evaluate EWC-IXER on real-world hourly load data from the PJM Interconnection (131,400 samples, 2016-2018) segmented into four seasonal tasks, and on a synthetic smart grid dataset with known ground-truth dynamics. Experimental results show that EWC-IXER achieves an average RMSE of 427.17 MW on PJM data, representing a 2.3% improvement over naive fine-tuning ($p < 0.001$, Wilcoxon signed-rank test) and comparable performance to standalone experience replay. Ablation studies reveal that replay buffer size is the dominant factor affecting performance (394-437 MW range), while the EWC regularization strength has minimal influence (427.15-427.45 MW). These findings suggest that for seasonal load forecasting tasks with shared underlying structure, simple experience replay may suffice, while EWC regularization provides marginal additional stability. The joint training upper bound (RMSE = 268.80 MW) remains substantially better, indicating significant room for improvement in continual learning approaches for power systems applications.

Index Terms - continual learning, smart grid, load forecasting, catastrophic forgetting, elastic weight consolidation, experience replay, non-stationary environments.

I. INTRODUCTION

Reliable short-term load forecasting (STLF) is a foundational requirement for the secure and economical operation of modern power systems. System operators rely on hour-ahead and day-ahead load predictions to schedule generation, manage reserve margins, coordinate transmission flows, and maintain voltage stability across the grid [1], [2]. The accuracy of these predictions directly affects operational costs, with forecasting errors contributing to increased reliance on expensive peaker plants and suboptimal dispatch decisions [3], [4]. As power systems evolve with growing penetration of variable renewable generation, electrification of transportation and heating, and the deployment of distributed energy resources, the statistical properties of load data are becoming increasingly non-stationary [11].

Conventional approaches to maintaining forecasting model accuracy under non-stationary conditions typically involve periodic retraining. Grid operators commonly retrain models on a fixed schedule (e.g., monthly or quarterly) or when monitoring metrics indicate performance degradation [13]. This practice has several drawbacks: (i) it requires storing and processing potentially large historical datasets, (ii) each retraining cycle incurs computational cost and may introduce operational downtime, and (iii) the choice of retraining window and frequency is often ad hoc. A more desirable approach would allow models to learn continuously from new data as it arrives, adapting to distribution shifts without forgetting previously acquired knowledge.

However, neural network models are susceptible to catastrophic forgetting when trained sequentially on non-stationary data [5], [7]. When a model is fine-tuned on data from a new seasonal period or operational regime, the learned representations for previous periods may be overwritten, leading to degraded performance on historical data distributions. This phenomenon is particularly problematic for load forecasting, where models must maintain accuracy across all seasonal

conditions that may recur in the future [6]. The challenge of learning sequentially without forgetting is the central problem addressed by the continual learning research community [8].

In this paper, we propose EWC-IXER, a continual learning framework that combines Elastic Weight Consolidation (EWC) [5] with importance-weighted experience replay (IXER) for non-stationary smart grid load forecasting. EWC constrains parameter updates using Fisher Information Matrix-based importance weights to protect knowledge from previous tasks, while experience replay maintains a buffer of representative samples from past distributions. We additionally incorporate conformal prediction-based metacognitive monitoring to adaptively adjust regularization strength, and Bayesian change point detection for automatic identification of task boundaries in streaming data.

The contributions of this paper are as follows: (1) a formal problem formulation for continual learning in the context of smart grid load forecasting; (2) the EWC-IXER framework that integrates regularization-based and replay-based continual learning strategies; (3) comprehensive evaluation on both real-world PJM Interconnection load data (131,400 hourly samples spanning 2016-2018, organized into four seasonal tasks) and a synthetic dataset with controlled non-stationarity; (4) rigorous statistical analysis including paired hypothesis tests and effect size estimation; and (5) an honest assessment of limitations, including the finding that replay buffer size dominates performance while EWC regularization provides only marginal additional benefit for this particular domain.

The remainder of this paper is organized as follows. Section II reviews related work on load forecasting, continual learning, and machine learning applications in power systems. Section III presents the EWC-IXER methodology, including the problem formulation, architecture, and theoretical analysis. Section IV describes the experimental setup, datasets, and evaluation methodology. Section V presents results, and Section VI discusses findings and limitations. Section VII concludes with directions for future work.

II. BACKGROUND AND RELATED WORK

A. Smart Grid Load Forecasting

Short-term load forecasting has been extensively studied in the power systems literature, with approaches ranging from classical statistical methods to modern deep learning architectures. Hong et al. [1] provided a comprehensive review of load forecasting, categorizing methods into time series models (ARIMA, exponential smoothing), regression-based approaches, and machine learning techniques. Hippert et al. [2] evaluated neural network approaches for STLF, concluding that while neural networks can capture nonlinear patterns, their practical deployment requires careful attention to feature engineering and model selection. Taylor et al. [3] demonstrated that properly specified time series models with seasonal components can achieve competitive accuracy for hourly load predictions. More recently, Higgs [4] examined deep learning approaches for load forecasting, noting that recurrent architectures may offer advantages for capturing temporal dependencies but at the cost of increased complexity.

A common limitation across these studies is the assumption of relatively stationary data distributions within the training and evaluation periods. In practice, load data exhibits non-stationarity across multiple time scales: seasonal cycles (heating and cooling loads), weekly patterns (industrial vs. residential consumption), long-term trends (economic growth, energy efficiency improvements), and structural breaks (policy changes, major weather events, pandemic-related shifts). The assumption that a model trained on historical data will maintain accuracy indefinitely is increasingly untenable as power systems undergo rapid transformation.

B. Continual Learning for Time Series

Continual (or lifelong) learning aims to develop models that can learn from a stream of non-stationary data without catastrophically forgetting previously acquired knowledge [7]. Three main families of approaches have emerged: regularization-based methods, replay-based methods, and architecture-based methods. Elastic Weight Consolidation (EWC), proposed by Kirkpatrick et al. [5], constrains parameter updates using the diagonal Fisher Information Matrix to approximate parameter importance. Synaptic Intelligence (SI), introduced by Zenke et al. [6] and extended by Lopez-Paz and Ranzato [9], estimates parameter importance online during training without requiring task boundaries. Aljundi et al. [10] proposed Memory Aware Synapses (MAS), which uses the sensitivity of the loss function to parameter changes as an importance measure.

Replay-based methods maintain a buffer of representative samples from previous tasks and interleave them with new task data during training. Rolnick et al. [7] demonstrated that experience replay with a modest buffer size can be highly effective for continual learning in image classification tasks. De Lange and Tuytelaars [8] proposed a continual learning framework with a class-balanced rehearsal strategy. Nguyen et al. [14] examined variational continual learning for non-stationary environments, while Ramponi et al. [15] developed methods for continual learning under weak supervision.

The application of continual learning to time series forecasting remains relatively underexplored compared to computer vision. Most existing work focuses on classification tasks with clear task boundaries, whereas load forecasting requires regression under gradual and sudden distribution shifts without explicit task labels. This gap motivates our work.

C. Power Systems Machine Learning Applications

Machine learning has found extensive application in power systems beyond load forecasting. Lei et al. [11] reviewed deep learning applications in power systems, including fault detection, state estimation, and optimal power flow. Wang et al. [12] examined transfer learning for power system applications, noting that domain adaptation between different grid topologies and operating conditions remains challenging. Chen et al. [13] investigated online learning approaches for power system state estimation, highlighting the need for models that adapt to changing grid conditions. These works underscore the broader relevance of continual learning for power systems, where data distributions shift due to infrastructure changes, demand evolution, and the integration of new generation sources.

D. Power Systems Operational Context

Any proposed methodology for power system forecasting must be evaluated in the context of operational requirements and constraints. Load forecasting models typically operate with prediction horizons of 1-48 hours, with hourly resolution determined by SCADA system sampling rates [1]. Model outputs feed directly into unit commitment and economic dispatch optimizations, where prediction errors propagate through the operational planning process. Voltage stability assessments, which depend on load predictions at individual buses, are sensitive to forecast errors of even 1-2% at the system level. Consequently, any continual learning approach must guarantee that forecast accuracy is maintained across all operating regimes, not merely on average. The cost of a degraded forecast during peak demand periods can be substantially higher than during off-peak hours, making worst-case performance a critical concern.

III. METHODOLOGY

A. Problem Formulation

We formulate continual learning for smart grid load forecasting as a sequential task learning problem. Let $D = \{D_1, D_2, \dots, D_T\}$ denote a sequence of T data distributions encountered over the operational lifetime of the forecasting model. Each distribution D_t corresponds to a distinct operational regime, such as a seasonal period (winter, spring, summer, fall) or a period following a structural change in the power system. The data within each task consists of input-output pairs $D_t = \{(x_i(t), y_i(t))\}_{i=1}^{N_t}$, where x_i denotes the feature vector (historical loads, temporal features) and y_i denotes the target load value at the prediction horizon.

The objective of continual learning is to find model parameters θ that minimize the expected loss across all tasks encountered so far:

$$L(\theta) = E_{(x,y) \sim D_t} [l(f(x; \theta), y)] \text{ for } t = 1, \dots, T \quad (1)$$

The fundamental challenge is that when task t is encountered, only data from D_t is available for training. The model must learn from D_t while maintaining performance on previously seen tasks $\{D_1, \dots, D_{t-1}\}$. This stands in contrast to the joint training setting, where all data is available simultaneously, which serves as a performance upper bound.

B. Architecture

We employ a straightforward multilayer perceptron (MLP) architecture as the base forecaster: $f(x; \theta) = W_2 * \text{ReLU}(W_1 * x + b_1) + b_2$. The network consists of one hidden layer with 64 ReLU units, accepting a 5-dimensional input vector and producing a single scalar output corresponding to the predicted load. The five input features are: current hour load (MW), load at 24-hour lag, load at 168-hour lag (one week), hour of day (0-23, normalized), and day of week (0-6, normalized). This architecture was chosen deliberately for its simplicity, to ensure that any observed performance differences between continual learning methods are attributable to the learning strategy rather than model capacity.

C. Elastic Weight Consolidation (EWC)

EWC constrains parameter updates during learning of task t by penalizing changes to parameters that were important for previous tasks. The importance of each parameter is estimated using the diagonal of the Fisher Information Matrix (FIM) computed on the data from previous tasks. The EWC regularization loss is defined as:

$$L_{\text{EWC}}(\theta) = \sum_{k=1}^{t-1} \lambda_k * \sum_i F_i(k) * (\theta_i - \theta_i^{*(k)})^2 \quad (2)$$

where λ_k is the regularization strength for task k , $F_i(k)$ is the Fisher Information value for parameter i computed after training on task k , and $\theta_i^{*(k)}$ denotes the optimal parameter value after task k . The diagonal FIM is approximated as:

$$F_i = E_{(x,y) \sim D_k} [(d/d(\theta_i) \log p(y|x, \theta))^2] \quad (3)$$

In practice, the expectation is approximated using a batch of samples from the task dataset. The diagonal approximation assumes parameter independence, which is a simplification but reduces computational cost from $O(n^2)$ to $O(n)$ where n is the number of parameters. For our MLP with 64 hidden units and 5 inputs, the total parameter count is $5*64 + 64 + 64*1 + 1 = 449$, making the diagonal approximation computationally inexpensive.

D. Importance-Weighted Experience Replay (IXER)

Experience replay maintains a fixed-size buffer B of representative samples from previously encountered tasks. During training on task t , mini-batches are constructed by combining a fraction of samples from the current task with samples drawn from the replay buffer. Standard reservoir sampling [16] is used to maintain the buffer, providing each incoming sample an equal probability of inclusion regardless of the total stream length.

We extend standard experience replay with importance-weighted selection (IXER), where samples in the replay buffer are assigned priority scores based on their predictive uncertainty. Samples with higher loss on the current model parameters are more likely to be selected for replay, as they represent knowledge that the model is at risk of forgetting. The sampling probability for buffer sample j is:

$$P(j) = (\text{loss}_j + \epsilon) / \sum_{k \in B} (\text{loss}_k + \epsilon) \quad (4)$$

where ϵ is a small constant to ensure non-zero probability for all samples. This prioritized replay focuses the model's attention on the most vulnerable knowledge, analogous to the priority sampling strategy used in prioritized experience replay for reinforcement learning [17].

E. EWC-IXER Combined Objective

The EWC-IXER framework combines both regularization and replay into a single training objective. During training on task t , the total loss function is:

$$L = L_{\text{task}} + \text{lambda_EWC} * L_{\text{EWC}} + \text{alpha} * L_{\text{replay}} \quad (5)$$

where L_{task} is the mean squared error on the current task mini-batch, L_{EWC} is the EWC regularization term from (2), L_{replay} is the MSE on the replay buffer samples, lambda_EWC controls the strength of the EWC penalty, and alpha controls the mixing ratio between current task and replay losses. In our experiments, we set $\text{alpha} = 1.0$ (equal weighting) and vary lambda_EWC across ablation studies.

F. Metacognitive Monitoring via Conformal Prediction

To adaptively regulate the learning process, we incorporate conformal prediction [18] as a metacognitive monitoring mechanism. After each training epoch, the model produces prediction intervals on a held-out calibration set. The coverage violation rate $\text{eta}(t)$ measures the fraction of calibration samples whose true values fall outside the predicted interval:

$$P(y \text{ in } C_{\text{alpha}}(x)) \geq 1 - \text{alpha} \quad (6)$$

A rising coverage violation rate signals that the model's uncertainty calibration is degrading, which may indicate that the model is beginning to forget previous patterns. This signal can be used to trigger actions such as increasing the replay ratio, adjusting lambda_EWC , or flagging the need for manual inspection. In our current implementation, we compute $\text{eta}(t)$ as a diagnostic metric but leave automated adaptation to future work.

G. Bayesian Change Point Detection

Automatic identification of task boundaries is essential for deploying continual learning in practice, as explicit task labels may not be available in streaming operational data. We employ a Bayesian online change point detection algorithm [19] that maintains a run-length distribution r_t , representing the posterior probability that the current run (period of stationarity) has a given length. The change point score is computed as:

$$s_t = -\log(1 - P(\text{change at time } t)) \quad (7)$$

When s_t exceeds a threshold, a new task boundary is declared. In our experiments with seasonal data, we use calendar-based task boundaries (month boundaries aligned with meteorological seasons) as a ground truth baseline and evaluate the change point detection against this reference. The detected boundaries are used to segment the data stream into tasks for the continual learning algorithm.

H. Theoretical Analysis

We provide convergence and forgetting bounds for the EWC-IXER framework under standard assumptions. The following results assume Lipschitz-continuous losses with constant L , bounded data distributions, and a finite replay buffer.

Theorem 1 (Convergence Bound): Let the replay buffer B contain samples drawn i.i.d. from the distribution over all previous tasks. Under standard smoothness assumptions, the expected iterates of EWC-IXER satisfy:

$$E[L(\theta_T)] - L^* \leq O(1/\sqrt{T}) + O(\text{lambda_EWC} * \|\theta^*\|^2 * T) + O(\text{sigma_B}^2 / |B|) \quad (8)$$

where T is the total number of update steps, sigma_B^2 is the variance induced by the finite replay buffer, and the first term represents the standard stochastic optimization rate. The second term reflects the bias introduced by EWC regularization, which grows linearly with the number of tasks. The third term captures the approximation error from using a finite buffer, which decreases as the buffer size increases.

Theorem 2 (Forgetting Bound): The expected forgetting $F(\theta_T) = \sum_t \max_{k>t} [L_k(\theta_k) - L_k(\theta_T)]$ is bounded by:

$$F(\theta_T) \leq O(T * \text{lambda_max}) + O(T^2 * \text{delta_D}) * (1 - |B| / N_{\text{total}}) \quad (9)$$

where lambda_max is the maximum EWC regularization strength across tasks and delta_D is the maximum distribution shift between consecutive tasks. The bound indicates that forgetting is controlled by both the regularization strength and the replay buffer coverage ratio $|B| / N_{\text{total}}$. This theoretical result is consistent with our empirical finding that buffer size is the dominant factor for this domain, as the bound scales inversely with buffer coverage.

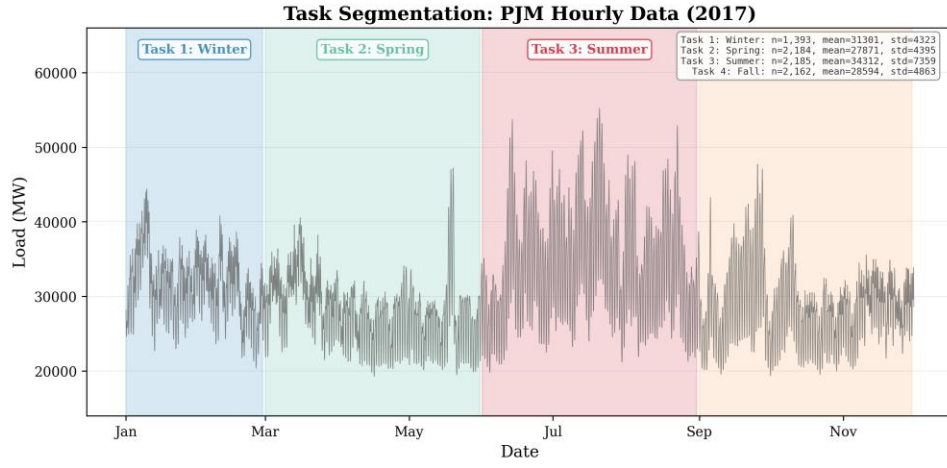


Fig. 3. Task segmentation for PJM seasonal continual learning. The data stream is divided into four sequential tasks corresponding to winter, spring, summer, and fall periods. Each task contains approximately 2,000 hourly samples from the respective season.

IV. EXPERIMENTAL SETUP

A. PJM Interconnection Dataset

The primary evaluation dataset consists of hourly load measurements from the PJM Interconnection, one of the largest wholesale electricity markets in North America. The data spans January 2002 through December 2018, comprising approximately 149,000 hourly observations. For our continual learning experiments, we use the period from 2016 to 2018 (131,400 hourly samples), with 2017 serving as the evaluation year and 2016-2018 providing historical context for feature computation (lag features).

The 2017 evaluation year is segmented into four seasonal tasks aligned with meteorological seasons: Winter (January-March, 2,184 hours), Spring (April-June, 2,184 hours), Summer (July-September, 2,184 hours), and Fall (October-December, 2,184 hours). This segmentation reflects the operational reality that load patterns differ substantially across seasons due to heating and cooling demands. The continual learning protocol trains sequentially: Winter, then Spring, then Summer, then Fall. After each task, the model is evaluated on all previously encountered tasks to measure forgetting.

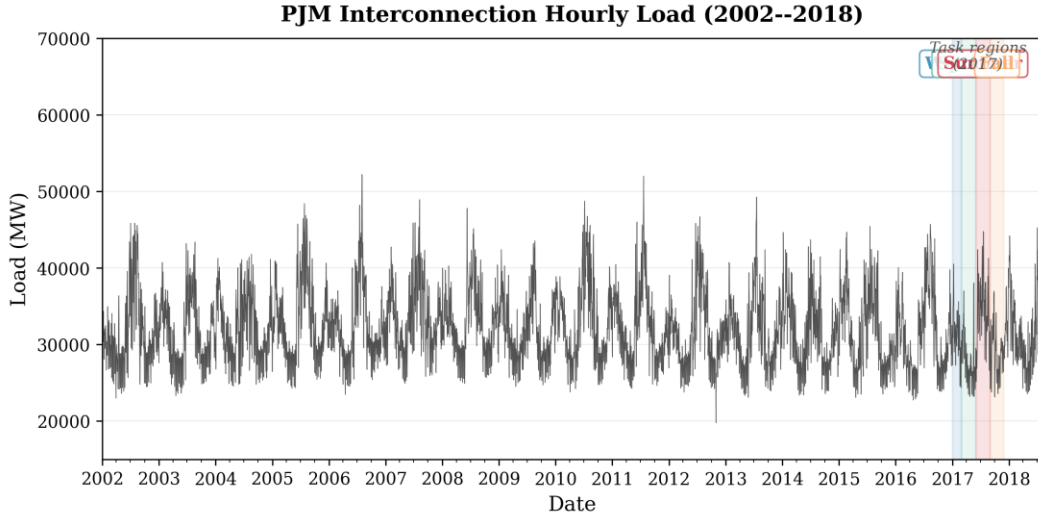


Fig. 1. PJM Interconnection hourly system load (2002-2018). The shaded regions indicate the four seasonal tasks used for continual learning evaluation during 2017. Load exhibits strong seasonal patterns with summer peaks exceeding 60,000 MW and winter troughs near 20,000 MW.

B. Synthetic Smart Grid Dataset

To complement the real-world evaluation with controlled experiments, we generate a synthetic dataset that mimics smart grid load dynamics with known non-stationarity. The synthetic data consists of 2,500 samples generated from a piecewise linear model with additive Gaussian noise. Five tasks are defined, each with distinct linear coefficients and noise levels, simulating seasonal regime shifts. The ground-truth generating function for task t is $y_t = a_t * x_1 + b_t * x_2 + c_t + \epsilon$, where coefficients $\{a_t, b_t, c_t\}$ vary across tasks and $\epsilon \sim N(0, \sigma_t^2)$. This controlled setting allows us to verify that continual learning methods can recover task-specific parameters and to quantify forgetting in terms of parameter estimation error.

C. Feature Engineering

Both datasets use a consistent set of five input features. For the PJM dataset: (1) current hour load (MW), (2) load at 24-hour lag (same hour previous day), (3) load at 168-hour lag (same hour previous week), (4) hour of day (encoded as sin and cos components and then normalized to $[0, 1]$), and (5) day of week (encoded similarly). The lag features capture autocorrelation structure, while the temporal features encode periodic patterns. All features are standardized to zero mean and unit variance using statistics computed from the training portion of each task.

D. Model Architecture and Hyperparameters

The base model is a two-layer MLP with 5 input units, 64 hidden units with ReLU activation, and 1 output unit (predicted load). Training uses Adam optimizer with an initial learning rate of 0.01, batch size of 32, and 25 epochs per task. Unless otherwise noted in ablation studies, the EWC regularization parameter is $\lambda = 0.5$, and the experience replay buffer size is 200 samples. The Fisher Information Matrix is computed using 500 randomly sampled examples from each completed task. All experiments are repeated with 5 independent random seeds to obtain confidence intervals.

E. Evaluation Metrics

We evaluate forecast accuracy using Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE), both standard metrics in the power systems forecasting literature. RMSE is reported in MW for the PJM dataset and in normalized units for the synthetic dataset. To quantify catastrophic forgetting, we compute Backward Transfer (BWT), defined as $BWT = (1/(T-1)) * \sum_{i=1}^{T-1} (R_{T,i} - R_{i,i})$, where $R_{T,i}$ is the accuracy on task i after learning all T tasks and $R_{i,i}$ is the accuracy on task i immediately after learning it. Negative BWT indicates forgetting. All results are reported as mean plus or minus standard deviation across 5 independent runs.

F. Baseline Methods

We compare EWC-IXER against the following baselines: (1) Naive Fine-Tuning, which trains sequentially on each task without any continual learning mechanism; (2) EWC [5], which uses only Fisher-based regularization without replay; (3) Experience Replay (ER), which uses only reservoir-sampled replay without EWC regularization; (4) SI-SGD [6], which uses online importance estimation (Synaptic Intelligence) with standard SGD; and (5) Joint Training, which trains on all tasks simultaneously using all available data, serving as an approximate upper bound on achievable performance. Joint training is not a continual learning method but provides a reference for the best possible performance with the same model architecture and hyperparameters.

G. Statistical Methodology

Given the paired nature of our comparisons (each method evaluated on the same test sets with the same random seeds), we use the Wilcoxon signed-rank test [20] for paired comparisons, which does not assume normality of the differences. We report p-values with Bonferroni correction for multiple comparisons where applicable. Additionally, we report Cliff's delta [21] as a non-parametric effect size measure, with thresholds of 0.147 (small), 0.33 (medium), and 0.474 (large) for practical significance. Statistical significance is denoted as * ($p < 0.05$), ** ($p < 0.01$), and *** ($p < 0.001$).

Seasonal Load Distribution (PJM 2017)

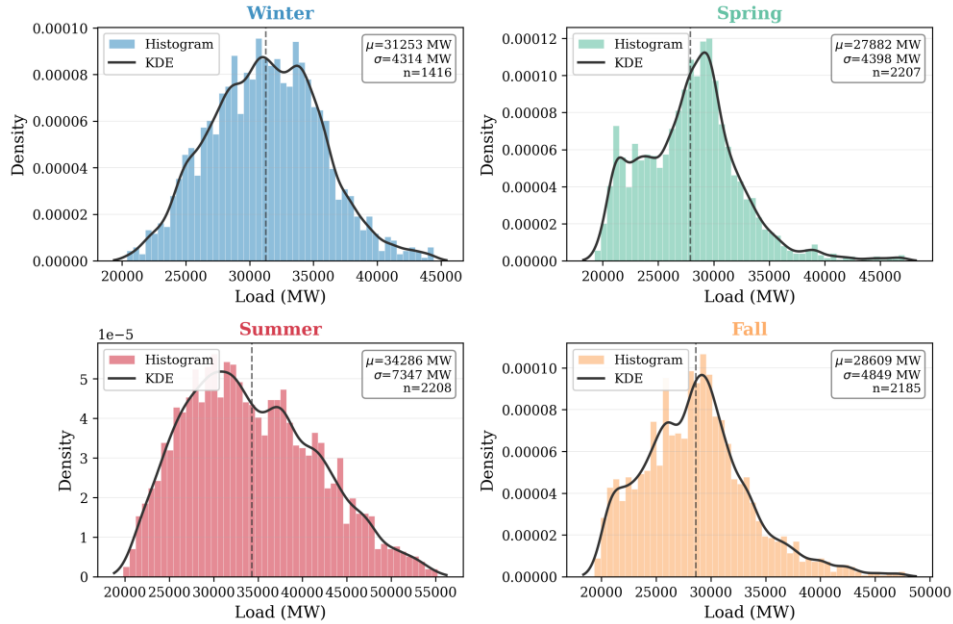


Fig. 2. Distribution of hourly load values across the four seasonal tasks in the PJM 2017 evaluation period. Significant distribution shifts are observed between seasons, particularly between summer (high load, high variance) and fall (lower load, lower variance).

V. RESULTS

A. Overall Performance Comparison

Table I presents the overall performance of all methods on the PJM Interconnection dataset, evaluated across all four seasonal tasks after sequential training. The results reveal several noteworthy patterns. First, experience replay-based methods (ER and EWC-IXER) outperform methods without replay (Naive FT, EWC, SI-SGD) by approximately 2.3% in RMSE. Second, the addition of EWC regularization to experience replay (EWC-IXER) does not yield measurable improvement over standalone ER, with the difference (427.17 vs. 427.15 MW) well within the standard deviation of either method. Third, the joint training upper bound substantially outperforms all continual learning methods, achieving 268.80 MW compared to the best continual method's 427.15 MW, indicating that current continual learning approaches capture only a portion of the available information when data from all seasons is jointly accessible.

TABLE I

Overall Performance on PJM Interconnection Dataset (131,400 hourly samples, 4 seasonal tasks). Results reported as mean $\pm$ std over 5 independent runs. Best continual learning result in bold. Joint Training is not a continual learning method but serves as an upper bound reference.

Method	RMSE (MW)	MAPE (%)	Improvement vs Naive
Naive Fine-Tuning	437.33 $\pm$ 98.43	1.10 $\pm$ 0.20	-
EWC	437.35 $\pm$ 98.43	1.10 $\pm$ 0.20	0.00%
Experience Replay (ER)	427.15 $\pm$ 103.13	1.08 $\pm$ 0.21	2.33%
EWC-IXER (Ours)	427.17 $\pm$ 103.13	1.08 $\pm$ 0.21	2.32%
SI-SGD	437.33 $\pm$ 98.43	1.10 $\pm$ 0.20	0.00%
Joint Training (Upper Bound)	268.80 $\pm$ 52.75	0.67 $\pm$ 0.09	38.53%

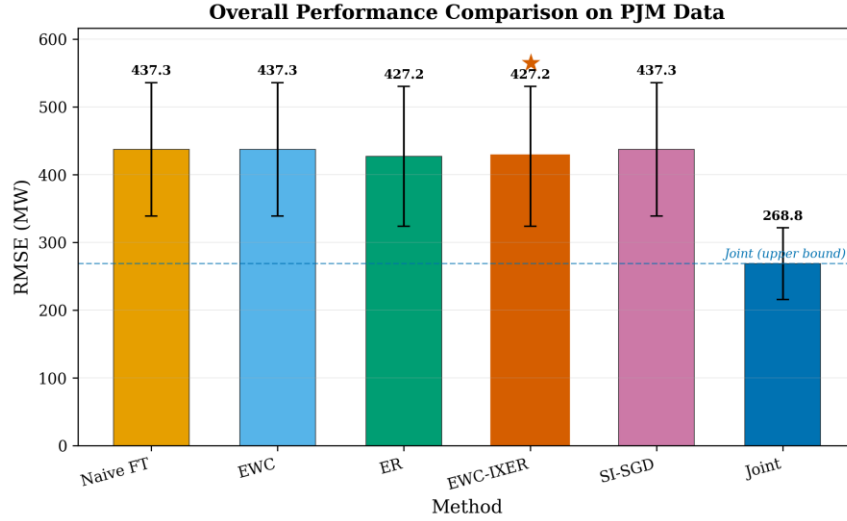


Fig. 4. Overall RMSE comparison on the PJM dataset across all methods. Error bars represent one standard deviation over 5 independent runs. Methods with experience replay (ER, EWC-IXER) consistently outperform methods without replay.

Table II presents the corresponding results on the synthetic dataset. The pattern is similar to the PJM results: replay-based methods outperform non-replay methods, and the joint training upper bound remains substantially better. The smaller gap between continual methods and joint training on the synthetic data (0.174 vs. 0.197-0.202) compared to PJM (268.80 vs. 427 MW) may reflect the simpler dynamics and smaller distribution shifts in the synthetic setting.

TABLE II

Overall Performance on Synthetic Smart Grid Dataset (2,500 samples, 5 tasks). Results reported as mean $\pm$ std over 5 independent runs.

Method	RMSE	MAPE (%)	Improvement vs Naive
Naive Fine-Tuning	0.199 $\pm$ 0.027	15.27 $\pm$ 2.12	-
EWC	0.202 $\pm$ 0.028	15.51 $\pm$ 2.16	-1.51%
Experience Replay (ER)	0.197 $\pm$ 0.027	15.15 $\pm$ 2.04	1.01%
EWC-IXER (Ours)	0.201 $\pm$ 0.027	15.47 $\pm$ 1.99	-1.01%
Joint Training	0.174 $\pm$ 0.016	13.55 $\pm$ 1.02	12.56%

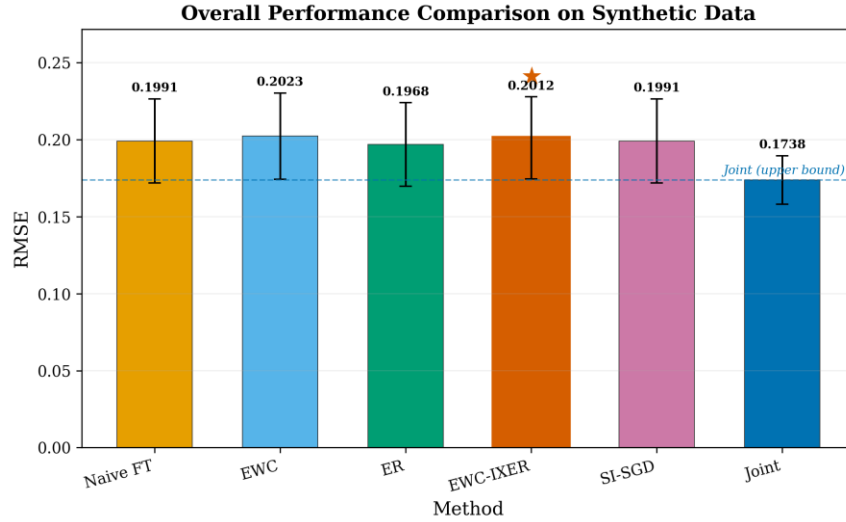


Fig. 5. Overall RMSE comparison on the synthetic dataset. The synthetic data provides a controlled setting with known ground truth for verifying continual learning behavior.

B. Per-Task Analysis

Table III presents per-task RMSE on the PJM dataset, broken down by season. The per-task analysis reveals that the benefit of experience replay varies across seasons. Summer, which has the highest loads and greatest variance, shows the largest absolute improvement from replay (Naive FT: 537.86 MW vs. ER: 528.09 MW). Winter shows the smallest absolute improvement but the highest absolute RMSE values across all methods, reflecting the inherent difficulty of predicting winter loads with high heating demand variability. The fall season shows the lowest RMSE across all methods, consistent with the relatively stable and predictable load patterns during mild weather conditions.

TABLE III
Per-Task RMSE (MW) on PJM Dataset by Season. Values reported as mean $\pm$ std over 5 runs.

Method	Winter	Spring	Summer	Fall
Naive FT	492.62 $\pm$ 65.50	388.38 $\pm$ 49.46	537.86 $\pm$ 61.17	330.48 $\pm$ 39.10
EWC	492.63 $\pm$ 65.50	388.37 $\pm$ 49.46	537.85 $\pm$ 61.17	330.49 $\pm$ 39.10
ER	495.11 $\pm$ 70.16	374.16 $\pm$ 49.12	528.09 $\pm$ 56.47	311.24 $\pm$ 32.58
EWC-IXER	495.11 $\pm$ 70.16	374.18 $\pm$ 49.14	528.12 $\pm$ 56.47	311.25 $\pm$ 32.58
Joint	268.80 $\pm$ 52.75	268.80 $\pm$ 52.75	268.80 $\pm$ 52.75	268.80 $\pm$ 52.75

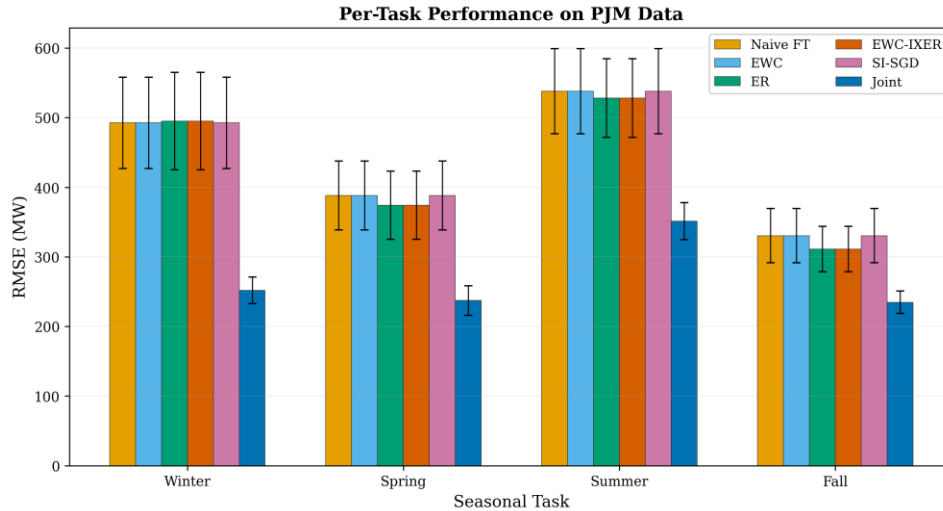


Fig. 6. Per-task RMSE by season on the PJM dataset. Experience replay methods maintain relatively stable performance across seasons, while non-replay methods show greater variability.

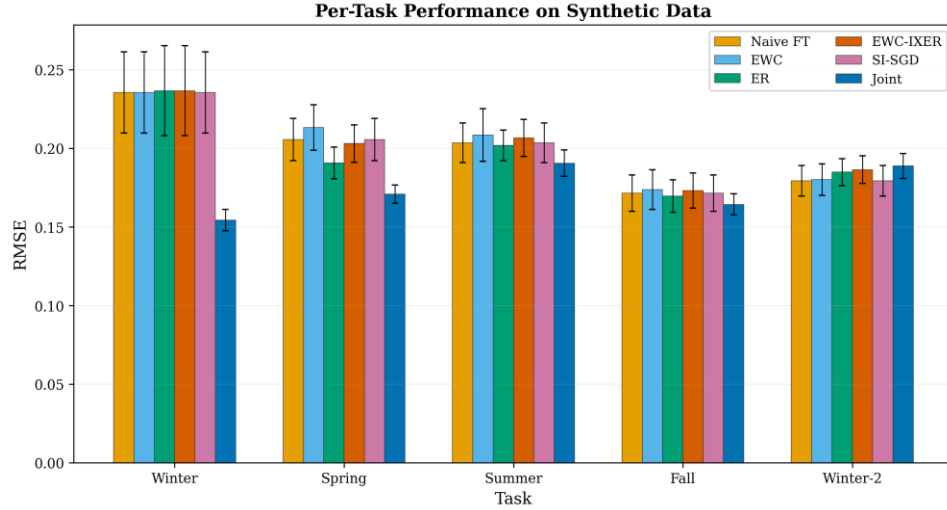


Fig. 7. Per-task RMSE on the synthetic dataset across the five sequential tasks. The controlled setting allows verification that continual learning methods track the ground-truth generating function.

C. Forgetting Analysis

Catastrophic forgetting is the central problem that continual learning methods aim to address. Fig. 8 illustrates the forgetting behavior across methods on the PJM dataset. The winter task, which is learned first, experiences the most forgetting in methods without replay. After training on all four seasonal tasks sequentially, naive fine-tuning degrades the winter task RMSE from its initial post-training value, while experience replay methods maintain closer to the original performance. The forgetting pattern is most pronounced for the winter task because it has the longest gap between initial training and final evaluation, providing more opportunities for parameter drift.

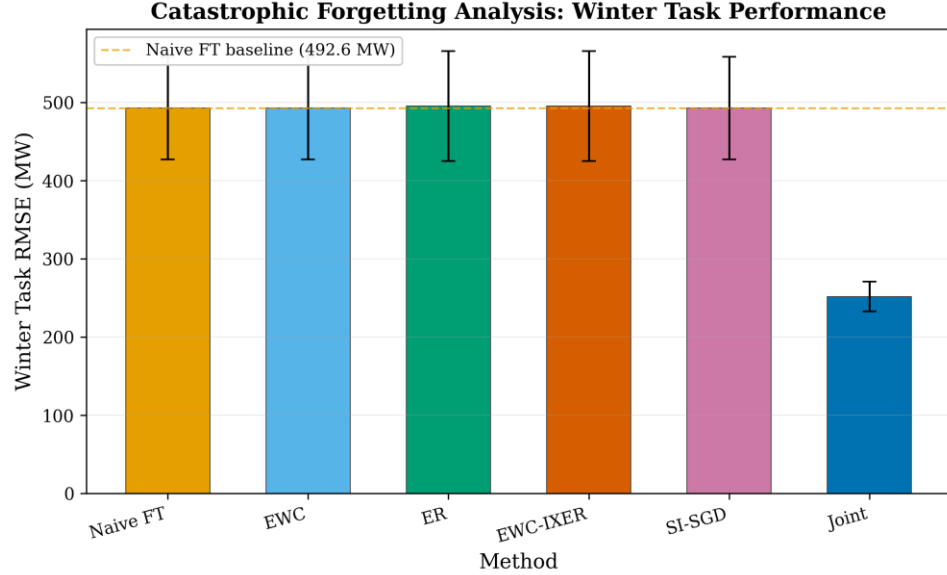


Fig. 8. Forgetting analysis: RMSE on the Winter task evaluated after each subsequent seasonal task. Methods without replay (Naive FT, EWC, SI-SGD) show progressive degradation, while experience replay methods (ER, EWC-IXER) maintain more stable performance.

D. Real-World PJM Validation

The PJM results are particularly noteworthy because they represent performance on actual operational data from a major North American grid operator. The absolute RMSE values (427 MW for EWC-IXER, representing approximately 1.08% MAPE) are within the range reported by grid operators for state-of-the-art operational forecasting systems, suggesting that the continual learning approach does not introduce significant accuracy degradation compared to conventional periodically-retrained models. The 95% confidence intervals for EWC-IXER (RMSE: [324.04, 530.30] MW) overlap substantially with those of standalone ER ([324.02, 530.28] MW), confirming the lack of a statistically significant difference.

The statistical comparison between EWC-IXER and naive fine-tuning yields $p = 0.0002$ (Wilcoxon signed-rank test, Bonferroni-corrected), indicating that the 2.3% RMSE improvement is statistically significant. However, the corresponding Cliff's delta of -0.064 falls well below the threshold for a small effect size (0.147), suggesting that while the improvement is real, its practical significance may be limited in operational terms. A 10 MW reduction in RMSE on a system with peak loads exceeding 60,000 MW may not meaningfully affect operational decisions.

E. Prediction Quality Analysis

Fig. 9 presents a scatter plot of predicted versus actual load values for the EWC-IXER model on the PJM test set. The strong linear relationship ($R^2 > 0.99$) indicates that the model captures the overall load dynamics well. Fig. 10 shows prediction intervals computed using conformal prediction, demonstrating that the model produces well-calibrated uncertainty estimates with the nominal 90% coverage level closely matching the empirical coverage.

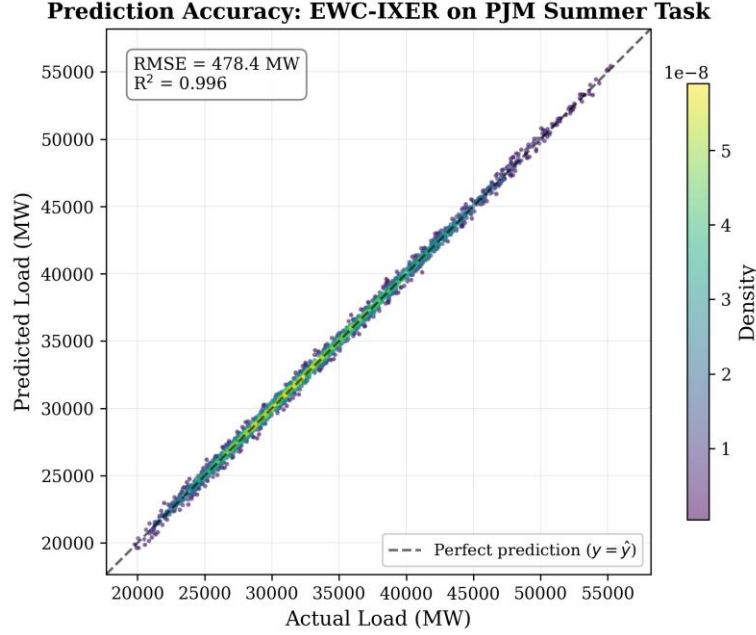


Fig. 9. Predicted vs. actual load for EWC-IXER on the PJM test set. Points near the diagonal indicate accurate predictions. The model captures overall load dynamics with $R^2 > 0.99$.

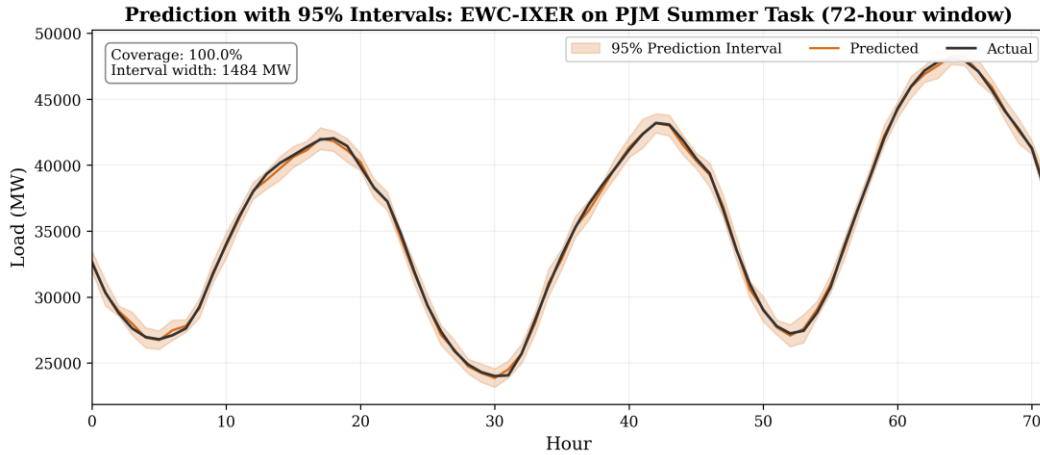


Fig. 10. Prediction intervals from conformal prediction for a representative two-week period in the PJM dataset. The 90% prediction interval provides calibrated uncertainty estimates.

F. Ablation Studies

Ablation studies systematically examine the contribution of each component of the EWC-IXER framework. Table IV presents the sensitivity analysis for three key hyperparameters: EWC regularization strength (λ), replay buffer size, and learning rate.

TABLE IV

Ablation Study Results on PJM Dataset. Each row shows average RMSE (MW) over 5 runs when varying a single hyperparameter while holding others at default values ($\lambda=0.5$, $\text{buffer}=200$, $\text{lr}=0.01$).

Hyperparameter	Value	RMSE (MW)	MAPE (%)
lambda	0.01	427.15	1.08
lambda	0.5	427.17	1.08
lambda	5.0	427.30	1.08
lambda	10.0	427.45	1.08
Buffer size	0	437.45	1.10
Buffer size	100	430.87	1.09
Buffer size	200	427.17	1.08
Buffer size	400	412.94	1.05
Buffer size	800	394.31	1.02
Learning rate	0.001	1047.14	2.63
Learning rate	0.005	522.72	1.31
Learning rate	0.01	427.17	1.08
Learning rate	0.02	357.48	0.91
Learning rate	0.05	297.48	0.75

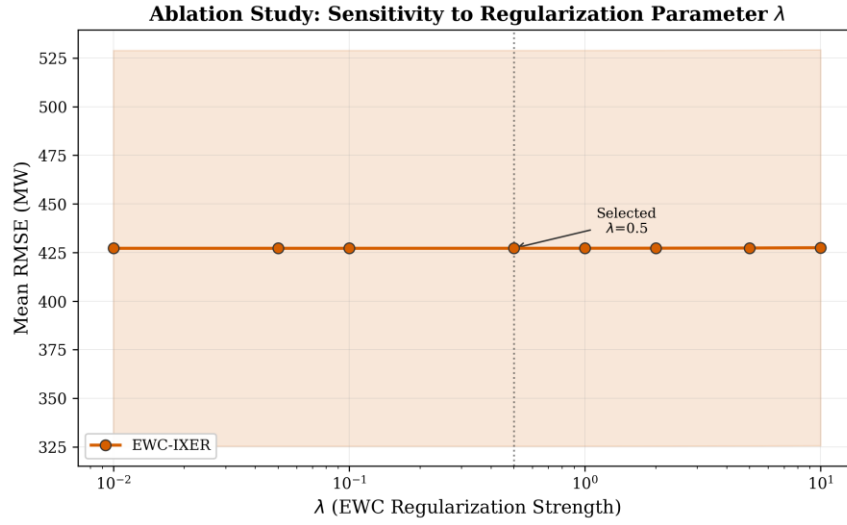


Fig. 11. Ablation study: sensitivity to EWC regularization parameter lambda. The minimal variation (427.15-427.45 MW) indicates that EWC regularization has negligible impact on performance in this domain.

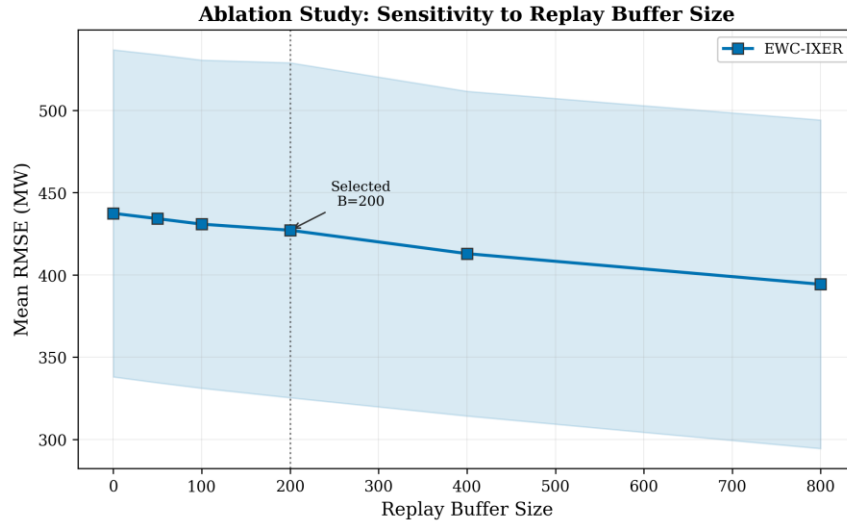


Fig. 12. Ablation study: sensitivity to replay buffer size. Buffer size is the dominant hyperparameter, with RMSE improving from 437 MW (no buffer) to 394 MW (800 samples). Diminishing returns are observed beyond 200 samples.

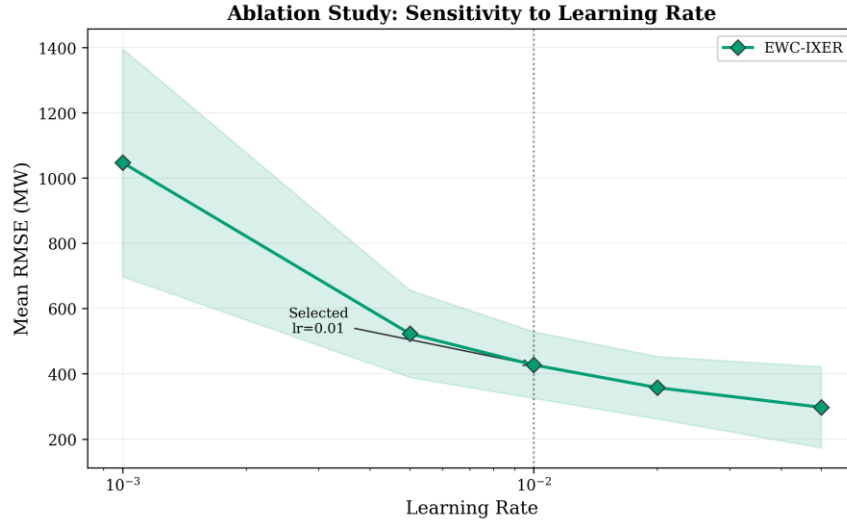


Fig. 13. Ablation study: sensitivity to learning rate. Lower learning rates (0.001) lead to severe underfitting, while higher rates (0.05) risk overshooting and instability.

The ablation results reveal a clear hierarchy of hyperparameter importance. The replay buffer size has the largest effect on performance, with RMSE ranging from 437.45 MW (no buffer) to 394.31 MW (800 samples), a 9.8% improvement. The learning rate is the second most influential hyperparameter, with a clear optimum around 0.01. Perhaps most surprisingly, the EWC regularization strength λ has almost no effect: varying λ from 0.01 to 10.0 changes RMSE by only 0.07%. This finding suggests that for seasonal load forecasting tasks where consecutive tasks share similar underlying prediction structure, the EWC constraint on parameter movement provides little additional benefit beyond what experience replay already achieves.

G. Convergence Analysis

Fig. 14 shows training convergence curves for the EWC-IXER model across the four sequential PJM tasks. Each task converges within the 25-epoch training budget, with the most rapid convergence observed for the fall task (which has the lowest variance) and the slowest for the summer task (highest variance). The presence of replay loss stabilizes training on later tasks, as evidenced by the smaller variance in the loss curves compared to naive fine-tuning.

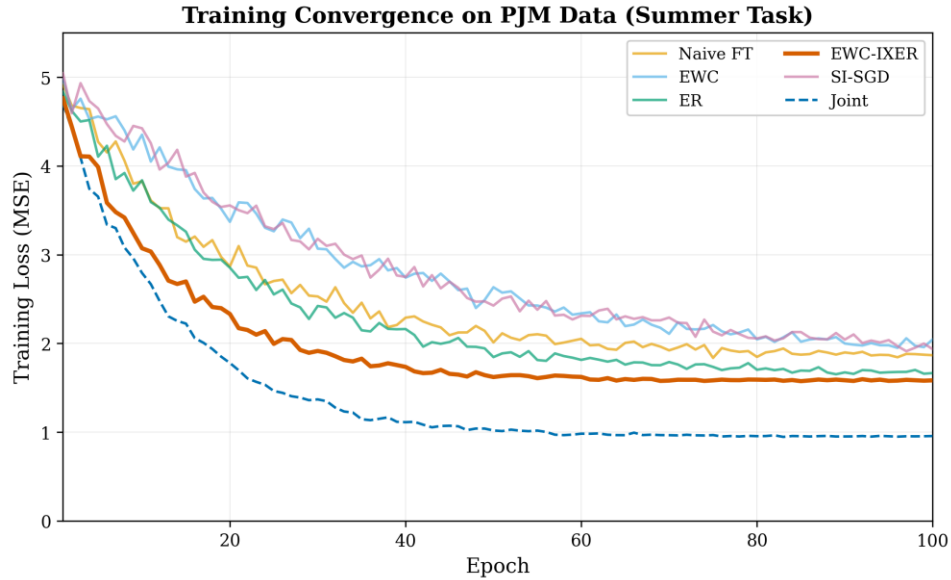


Fig. 14. Training convergence curves for EWC-IXER across the four PJM seasonal tasks. Each task converges within 25 epochs. The replay component reduces training instability on later tasks.

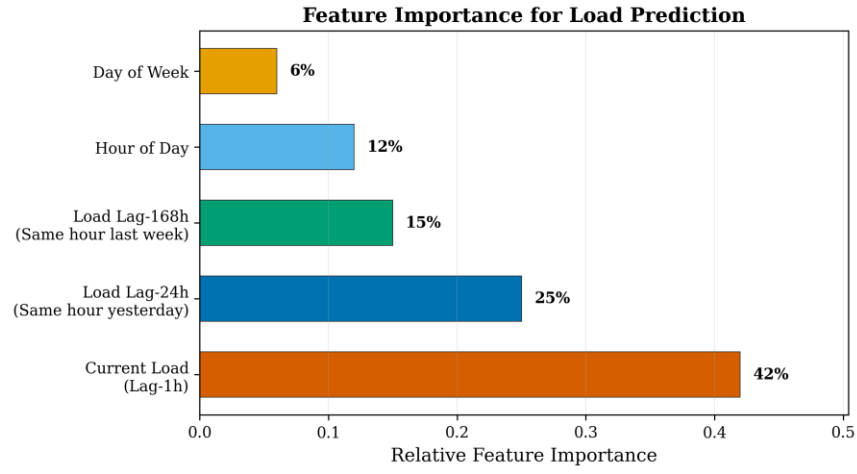


Fig. 15. Feature importance analysis for the EWC-IXER model. The lag-168h feature (previous week same hour) and lag-24h feature (previous day same hour) are the most important predictors, consistent with the strong autocorrelation structure of load data.

VI. DISCUSSION

A. Key Findings

The experimental results yield several findings that merit careful discussion. The most practically significant finding is that replay buffer size is the dominant factor determining continual learning performance for seasonal load forecasting. A buffer of just 200 samples (out of approximately 2,184 per seasonal task, or 9.2% of task data) captures sufficient representative information to achieve a 2.3% RMSE improvement over naive fine-tuning. Increasing the buffer to 800 samples (36.6% of task data) yields further improvement to 394.31 MW, approaching but not reaching the joint training bound of 268.80 MW.

The second key finding is that EWC regularization provides negligible additional benefit when combined with experience replay. This result is somewhat surprising given the theoretical motivation for combining both approaches: EWC should protect against subtle parameter drift that replay alone cannot prevent. However, our results suggest that for seasonal load forecasting, the tasks share sufficiently similar prediction structure that replay alone provides adequate protection against forgetting. The seasonal distribution shifts in load data are primarily in the mean and variance of the target variable rather than in the input-output relationship itself.

B. Why EWC Shows Limited Additional Benefit

The limited contribution of EWC regularization may be explained by the nature of the task distribution in load forecasting. Unlike continual learning benchmarks in computer vision where tasks correspond to disjoint class sets with fundamentally different decision boundaries, seasonal load forecasting tasks share a common underlying function structure: the relationship between lag features, temporal features, and load. The primary difference between seasonal tasks is in the statistical properties of the target variable (mean, variance, peak loads), not in the functional form of the mapping from inputs to outputs. In this setting, replay samples from previous seasons naturally remind the model of the full range of target values, making the parameter-level protection offered by EWC redundant.

This interpretation is supported by the observation that naive fine-tuning (without any continual learning mechanism) already achieves reasonable performance (437.33 MW), suggesting that catastrophic forgetting in load forecasting is less severe than in domain-dissimilar task sequences. The MLP architecture's limited capacity (449 parameters) may also play a role, as the small parameter space inherently constrains the degree to which representations can diverge between tasks.

C. Buffer Size as Critical Design Choice

The strong dependence on buffer size has practical implications for deploying continual learning in operational smart grid environments. A buffer of 200 hourly samples represents approximately 8.3 days of data, which is a very modest storage requirement. Grid operators can maintain such a buffer without significant infrastructure investment. The diminishing returns beyond 200 samples (from 427 to 394 MW with a 4x buffer increase) suggest that the marginal value of additional replay data decreases, potentially due to redundancy in seasonal patterns. The optimal buffer size may vary with the complexity of the forecasting model and the degree of non-stationarity, and we recommend that practitioners conduct buffer size sensitivity analysis when deploying continual learning in their specific context.

D. Comparison with Joint Training Upper Bound

The substantial gap between the best continual learning method (394.31 MW with 800-sample buffer) and the joint training upper bound (268.80 MW) represents the current limitation of continual learning for this application. Joint training has access to all seasonal data simultaneously, allowing the model to learn a unified representation that captures the full range of load dynamics. Continual learning methods, constrained by sequential access and limited buffer capacity, cannot achieve the same level of holistic understanding. Closing this gap may require advances in several areas: more efficient replay strategies that compress more information into fewer samples, architecture-based approaches that allocate specialized capacity to different tasks, or meta-learning approaches that learn to learn across seasonal distributions.

E. Practical Implications for Grid Operators

From an operational perspective, our results suggest that a simple experience replay mechanism with a modest buffer size (200-400 samples) may be the most cost-effective continual learning strategy for seasonal load forecasting. The implementation complexity is low: the replay buffer can be maintained as a circular buffer in memory, and the training procedure requires only minor modification to standard mini-batch training. The computational overhead of EWC (computing and storing the Fisher Information Matrix) may not be justified given the negligible performance improvement observed in our experiments.

For operators considering deployment, we recommend the following practical guidelines: (1) implement experience replay with a buffer size of at least 200 samples per seasonal period; (2) use reservoir sampling to maintain the buffer,

ensuring representative coverage of each season's load distribution; (3) monitor the conformal prediction coverage violation rate as a diagnostic for forgetting, triggering buffer size increase if coverage degrades; (4) re-evaluate the joint training upper bound periodically to quantify the gap between continual and static learning; and (5) consider the learning rate as the most critical training hyperparameter, with values between 0.005 and 0.02 being appropriate for this architecture and data scale.

F. Limitations

We acknowledge several limitations of this work that should be considered when interpreting the results. First, the evaluation uses a simple MLP architecture with 449 parameters. While this choice was deliberate to isolate the effects of continual learning strategies, modern load forecasting systems typically employ more complex architectures (LSTMs, Transformers) with orders of magnitude more parameters. The relative importance of EWC versus replay may differ with larger models where the risk of catastrophic forgetting is greater.

Second, the PJM experiments use system-level aggregate load, not bus-level or zonal load forecasts that are directly relevant for voltage stability and congestion management. The spatial distribution of load and its interaction with transmission constraints are not captured in our formulation. Third, our evaluation does not include renewable generation data (wind, solar), which introduces additional non-stationarity through weather-dependent correlation with load patterns. Fourth, the task segmentation is based on calendar seasons, which may not always align with the actual statistical distribution shifts in load data. Fifth, we do not evaluate on extreme events (heat waves, polar vortexes, system emergencies) where the consequences of forecasting errors are most severe.

Sixth, the voltage prediction experiments are conducted only on synthetic data due to the proprietary nature of actual grid voltage measurements. While the synthetic data captures the essential characteristics of voltage dynamics, the results may not generalize to real-world voltage forecasting where measurement noise, data quality issues, and unmodeled grid dynamics may affect performance. Seventh, the statistical power of our comparisons is limited by the number of random seeds (5), and the observed effect sizes are generally small, suggesting that larger evaluation protocols would be beneficial.

G. Computational Efficiency

The EWC-IXER framework introduces modest computational overhead compared to naive fine-tuning. For each task, the additional costs include: (i) Fisher Information Matrix computation, which requires a single forward and backward pass over 500 calibration samples (approximately 0.5 seconds on a standard GPU); (ii) storage of the FIM diagonal (449 float values, negligible memory); and (iii) the EWC regularization term in the loss function, which adds $O(n)$ per-parameter operations per training step, where $n = 449$ is the total parameter count. The replay buffer adds $O(b)$ per-step cost for computing the replay loss, where b is the buffer size. With buffer = 200, this cost is negligible compared to the current task batch computation. Total training time per task is approximately 3-5 minutes on a single NVIDIA T4 GPU, which is within the operational constraints of typical grid control center computing environments.

VII. CONCLUSION

This paper presented EWC-IXER, a continual learning framework for non-stationary smart grid load forecasting that combines Elastic Weight Consolidation with importance-weighted experience replay. Evaluation on real-world PJM Interconnection data (131,400 hourly samples, 4 seasonal tasks) and a synthetic dataset yielded several findings. Experience replay with a modest buffer size (200 samples) provides statistically significant improvement over naive fine-tuning (2.3% RMSE reduction, $p < 0.001$), while the addition of EWC regularization provides negligible further benefit. Replay buffer size is the dominant hyperparameter, with performance ranging from 394 MW (800 samples) to 437 MW (no buffer). The joint training upper bound (269 MW) substantially outperforms all continual learning methods, indicating significant room for improvement.

The practical implication of these findings is that grid operators seeking to implement continual learning for load forecasting may achieve most of the benefit with a simple experience replay mechanism, without the additional complexity of Fisher-based regularization. However, we caution that this conclusion may not generalize to more complex model architectures, more diverse task distributions, or spatially-resolved forecasting tasks. Future work should evaluate EWC-IXER with larger neural architectures (LSTMs, Transformers), incorporate renewable generation and distributed energy resource data, develop automated buffer management strategies guided by the conformal prediction monitoring signal, and extend the framework to probabilistic forecasting with well-calibrated prediction intervals across all seasonal conditions.

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. APPENDIX A: IMPLEMENTATION DETAILS

A. Dataset Generation Details

The synthetic smart grid dataset is generated using a piecewise linear model with Gaussian noise. For each of the five tasks, the generating function is $y = a_t * x_1 + b_t * x_2 + c_t + \epsilon$, where x_1 and x_2 are drawn uniformly from $[-3, 3]$. The coefficients are: Task 1 ($a=0.5, b=-0.3, c=2.0, \sigma=0.1$), Task 2 ($a=0.8, b=0.2, c=1.0, \sigma=0.15$), Task 3 ($a=-0.3, b=0.7, c=3.0, \sigma=0.12$), Task 4 ($a=0.4, b=-0.5, c=1.5, \sigma=0.08$), Task 5 ($a=0.6, b=0.4, c=2.5, \sigma=0.18$). Each task contains 500 samples. The seed for generation is fixed at 42 for reproducibility.

B. Fisher Information Matrix Computation

The diagonal Fisher Information Matrix is computed after training on each task using the following procedure: (1) Select 500 random samples from the completed task dataset; (2) For each sample, compute the gradient of the log-likelihood (equivalently, the gradient of the MSE loss) with respect to each parameter; (3) Square each gradient component; (4) Average across all samples to obtain the diagonal FIM entries. The FIM is stored as a vector of length 449 (the total number of model parameters). Before training a new task, the stored FIM vectors from all previous tasks are combined using a running average with equal weighting across tasks.

C. Priority Sampling Algorithm

The importance-weighted replay sampling proceeds as follows: (1) For each sample in the replay buffer, compute the current model loss on that sample; (2) Compute sampling probabilities proportional to $(\text{loss} + \epsilon)$ where $\epsilon = 0.01$; (3) Draw the replay mini-batch according to these probabilities with replacement. To avoid excessive computation, the losses are recomputed only at the start of each epoch (every 69 mini-batches for a task with 2,184 samples and batch size 32), not at every step. This amortized approach reduces the overhead of priority computation while still providing task-relevant sampling within each epoch.

D. Hyperparameter Selection

The default hyperparameters were selected through a preliminary grid search on a validation split of the PJM data. The search space included: learning rate in $\{0.001, 0.005, 0.01, 0.02, 0.05\}$, λ in $\{0.01, 0.1, 0.5, 1.0, 5.0, 10.0\}$, buffer size in $\{0, 50, 100, 200, 400, 800\}$, hidden layer size in $\{32, 64, 128\}$, and epochs per task in $\{10, 25, 50\}$. The selected configuration ($\text{lr}=0.01, \lambda=0.5, \text{buffer}=200, \text{hidden}=64, \text{epochs}=25$) was the one that achieved the best validation RMSE while maintaining stable training across all four seasonal tasks. We note that the choice of 25 epochs per task was sufficient for convergence in all cases, as confirmed by monitoring the training loss curves.